

PROGRAM 21:

```
vedsk0831@DESKTOP-83FU7HB:~$ cd ~
cd OSSP
mkdir -p Program22
cd Program22
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program22$ pwd
/home/vedsk0831/OSSP/Program22
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program22$ nano program22.c
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program22$ gcc -Wall -Wextra -g program22.c -o program22
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program22$ ls -l
total 24
-rwxr-xr-x 1 vedsk0831 vedsk0831 18152 Oct  6 14:00 program22
-rw-r--r-- 1 vedsk0831 vedsk0831 1583 Oct  6 14:00 program22.c
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program22$ ./program22
=====
OSSP PROGRAM 22
Valgrind and Automated Testing
=====

Test 1: Memory allocation
Buffer: OSSP Program 22
Memory released successfully

Test 2: Multiple allocations
Numbers: 1 2 3 4 5
Memory released successfully

Test 3: String handling
Memory testing with Valgrind
String memory released

All tests completed.
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program22$ valgrind --version
valgrind-3.26.0
```

```
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program22$ valgrind --leak-check=full --show-leak-kinds=all ./program22
==2260== Memcheck, a memory error detector
==2260== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2260== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2260== Command: ./program22
==2260==
=====
OSSP PROGRAM 22
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Test 1: Memory allocation
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All tests completed.
==2260==
==2260== HEAP SUMMARY:
==2260==    in use at exit: 0 bytes in 0 blocks
==2260==   total heap usage: 4 allocs, 4 frees, 1,194 bytes allocated
==2260==
==2260== All heap blocks were freed -- no leaks are possible
==2260==
==2260== For lists of detected and suppressed errors, rerun with: -s
==2260== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

```

Test 2: Running program
PASS: Program executed successfully

Test 3: Checking output
PASS: Expected output found

Test 4: Checking memory
PASS: No memory leak detected

=====
TESTING COMPLETED
=====
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program22$ cat output.txt
=====
OSSP PROGRAM 22
Valgrind and Automated Testing
=====

Test 1: Memory allocation
Buffer: OSSP Program 22
Memory released successfully

Test 2: Multiple allocations
Numbers: 1 2 3 4 5
Memory released successfully

Test 3: String handling
Memory testing with Valgrind
String memory released

All tests completed.
vedsk0831@DESKTOP-83FU7HB:~/OSSP/Program22$ |

```

CODE:

```
#!/bin/bash
```

```
echo "====="
```

```
echo " PROGRAM 22 AUTOMATED TEST"
```

```
echo "====="
```

```
echo ""
```

```
echo "Test 1: Checking executable"
```

```
if [ -f "./program22" ]; then
```

```
    echo "PASS: Executable exists"
else
    echo "FAIL: Executable not found"
    exit 1
fi
```

```
echo ""
echo "Test 2: Running program"
```

```
./program22 > output.txt
```

```
if [ $? -eq 0 ]; then
    echo "PASS: Program executed successfully"
else
    echo "FAIL: Program execution failed"
    exit 1
fi
```

```
echo ""
echo "Test 3: Checking output"
```

```
if grep -q "All tests completed." output.txt; then
    echo "PASS: Expected output found"
else
    echo "FAIL: Expected output missing"
```

```
        exit 1
    fi

    echo ""
    echo "Test 4: Checking memory"

    valgrind --leak-check=full ./program22 > valgrind.txt 2>&1

    if grep -q "All heap blocks were freed" valgrind.txt; then
        echo "PASS: No memory leak detected"
    else
        echo "FAIL: Memory leak may exist"
    fi

    echo ""
    echo "=====
    echo " TESTING COMPLETED"
    echo "=====
```